Stationary Strings Theorem VI: Dirac–Majorana Chiral Peeling in Navier–Stokes Vortex Blowup, the Λ_{Meral} Baryogenesis Constant, and CERN ALPHA / 511 keV Observational Synthesis

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Abstract

Continuing the Theorems I–V series, this 6th study couples the September 2026 Navier–Stokes–FENE-P inner-core vortex dynamics to relativistic Dirac-Weyl spinor fields (ψ_L, ψ_R):

1. Under core frequency freezing $\omega_D(s) \to 0$ and string-tension-induced Majorana mass $M_{\text{Maj}}(X, s)$, chiral peeling occurs ($Q_5(30) = -737.97$, flip fraction 0.6440, Majorana on/off difference 1649.08, Rabi cross-check error 7.22×10^{-15} — **Test T13**).
2. From the electroweak sphaleron coefficient $\frac{28}{79}$ and the inner-core helicity integral, a new dimensionless universal constant, the **Meral Constant** $\Lambda_{\text{Meral}}(h)$, and its $h \to 0^+$ closed analytic form ($\Lambda^{(0)}_{\text{Meral}} = 0.07226943267767906$, difference vs. `quad` 0.0) are derived, locking the Planck CMB baryon asymmetry ($\eta_B = 6.12 \times 10^{-10}$) at $\epsilon^*_{\text{CP}} = 3.044224786045 \times 10^{-11}$ (**Test T14**).
3. In the low-shear limit, CERN ALPHA-g ($a_{\bar{g}}/g = 1.0$, 1.21σ marginal agreement with measurement), CERN ALPHA 2026 1.03/1.07 T Breit–Rabi hyperfine splitting ($f_{da} - f_{cb} = a_{1S}/h$ field-independence at $\sim 10^{-9}$ kHz, SME bound $\kappa_{a5} \lesssim 7.0 \times 10^{-7}$) and the 511 keV + ortho-positronium ring-fuel spectrum (**Test T15**) are verified.

1 Introduction and Model–Source Separation (Honesty Framework)

In Theorems I–V [1–5], the inner-core vortex singularity of the September 2026 Navier–Stokes solution [6] was shown to be arrested by elastic string tension, axis-regularized, and brought to self-consistent saturation at full PDE level. This work shows that the fermionic counterpart of the "Stationary String ($\omega_n \to 0$)" phase transition in vortex shear is **Dirac → Majorana chiral peeling** (`arXiv:2610.10688` [7]), and derives the dimensionless constant linking cosmological matter-antimatter asymmetry.

Model–Source Separation: Spinor identities (ψ_L matter / ψ_R counterpart modes), the $M_{\rm Maj}$ singlet-tension coupling, the Boltzmann S/W ansatz, the SME κ_{a5} benchmark and the Villata [8] narrative are the *theoretical model construction* of this study; CERN ALPHA-g [9], ALPHA 2026 4 ppm [10], BASE 2026 [11], Siegert–Yoneda 511 keV [12] and Planck 2018 ($\eta_B^{\rm obs} = 6.12 \times 10^{-10}$) numbers are directly verified *experimental/observational data*.

2 Task 8 / Test T13 — Dirac–Majorana Flip-Flop and Chiral Peeling in Curved/Vortex Space

The four-component Dirac spinor $\psi = (\psi_L, \psi_R)^T$ enters spin-vortex (chiral vortical) coupling with the axial vorticity of the Navier–Stokes inner-core vortex. On similarity time $s = -\ln\tau$ and co-moving coordinate X, the spinor amplitudes obey (4-real Radau system):

$$\begin{aligned}
\frac{d\psi_L(X,s)}{ds} &= -\frac{1}{2}\Omega_{\rm vort}(X,s)\psi_L(X,s) - i\,\omega_D(X,s)\psi_R(X,s), \\
\frac{d\psi_R(X,s)}{ds} &= +\frac{1}{2}\Omega_{\rm vort}(X,s)\psi_R(X,s) - i\,\omega_D(X,s)\psi_L(X,s) - i\,M_{\rm Maj}(X,s)\psi_R^*(X,s),
\end{aligned} \tag{1}$$

with $\omega_D(X,s) = \omega_{D,0}\, e^{-hs}\, F(X,s)/F_0 \xrightarrow{s\to\infty} 0$ (frequency freezing),

$$M_{\rm Maj}(X,s) = \lambda_{\rm Maj}\,\mu_s\,\mathcal{K}_{\rm curv}(X)\, e^{-(1+h)s}\left(\tilde{\ell}(X,s)^2 - \ell_0^2\right)\phi_{\rm FENE}, \tag{2}$$

and chiral vortex potential $\Omega_{\rm vort}(X,s) = \kappa_{\rm chiral}\, F(X,s)\, U_{\rm full}(X,0)\, \hat{G}(X)\, e^{-hs}$ ($\phi_{\rm FENE} = \max(1 - \tilde{\ell}^2/b_{\max}, 0)$ FENE saturation cap).

Here the chiral charge is $Q_5 = |\psi_L|^2 - |\psi_R|^2$ (peeling points $Q_5 < 0$). The $\pm\Omega_{\rm vort}/2$ chiral pumping drives the norm exponentially by open-system construction ($R^2 \sim 10^3$ in the core); in the closed $\Omega = M = 0$ limit unitary Rabi oscillation is preserved at 7.22×10^{-15} machine precision.

Table 1: **Test T13** (Table 11): Dirac–Majorana flip-flop chiral peeling ($h = 0.005, \mu_s = 0.1, s = 30$).

X	$Q_5(30)$	flip fraction	Majorana on/off difference
0.5	1.3955838270097018	0.4890594430379023	31.294628543170404
1.0	−737.9704871653987	0.6440203291385183	1649.0783062634898
2.0	1.0074527715918244	0.038061022778089405	0.05869937428714955

3 Task 9 / Test T14 — Derivation, Closed Form and Boltzmann Freeze-Out of the Universal Λ_{Meral} Vortex-Baryogenesis Constant

For $N_g = 3$ generations and $N_H = 1$ Higgs doublet, the electroweak sphaleron conversion coefficient is $C_{\text{sph}} = \frac{8N_g+4N_H}{22N_g+13N_H} = \frac{28}{79} = 0.35443037974683544$.

Theorem 1 (Meral Constant and Closed Analytic Form): *The dimensionless universal vortex-chirality integral*

$$\Lambda_{\text{Meral}}(h) = \frac{28}{79}\int_0^\infty \mathcal{K}_{\text{curv}}(X)\, F_{\text{base}}(X,0)\, U_{\text{full}}(X,0)\, \sqrt{a(X,0)^2 + \beta_0^2 l(X,0)^2}\, dX \tag{3}$$

(absolutely convergent, $\sim X^{-3-2h}$ tail) is solved exactly in the $h \to 0^+$ inviscid limit ($F_{\text{base}} = (1+X)^{-1}$, $U_{\text{full}} = \frac{1}{2}(1+X)^{-1}$, $a = \frac{2X}{1+X}$, $l = \frac{1}{1+X}$, $\beta_0 = 1/2$, $\mathcal{K}_{\text{curv}} = \frac{1}{1+X}$) by Euler t-substitution ($\sqrt{4X^2+1/4} = 2X + t$) and partial fractions:

$$\boxed{\Lambda^{(0)}_{\text{Meral}} = \frac{28}{79}\cdot\frac{1}{2}\left[\frac{163}{578} + \frac{64\sqrt{17}}{4913}\ln\left(\frac{21+5\sqrt{17}}{4}\right)\right] = 0.07226943267767906} \tag{4}$$

(Logarithm arguments are positive; difference vs. `quad` *is* 0.0*.)* □

The coupled Boltzmann–vortex equation $\frac{dY_{B-L}}{ds} = \epsilon_{\text{CP}}\,\mathcal{S}_{\text{vort}}(s) - \Delta W(s)\, Y_{B-L}(s)$ ($\mathcal{S}$: normalized string-tension rate, $\Delta W = e^{-s}$ decaying washout; $\eta_B = 7.04 \times \frac{28}{79} Y_{B-L}$) solved over $s \in [0, 60]$ locks the Planck value at $\epsilon^*_{\text{CP}} = 3.04422478604457 \times 10^{-11}$ (back-check relative error 3.26×10^{-13}).

Table 2: **Test T14** (Table 12): Meral constant, closed form and Boltzmann baryogenesis.

h	$\Lambda_{\text{Meral}}(h)$
0.002	0.07216110395029413
0.005	0.07199925393857878
0.008	0.07183817042047706
0.010	0.07173120437432301
0 (quad / closed form)	0.07226943267767906
$\epsilon^*_{\text{CP}} = 3.04422478604457 \times 10^{-11}$, η back-check 3.26×10^{-13}	

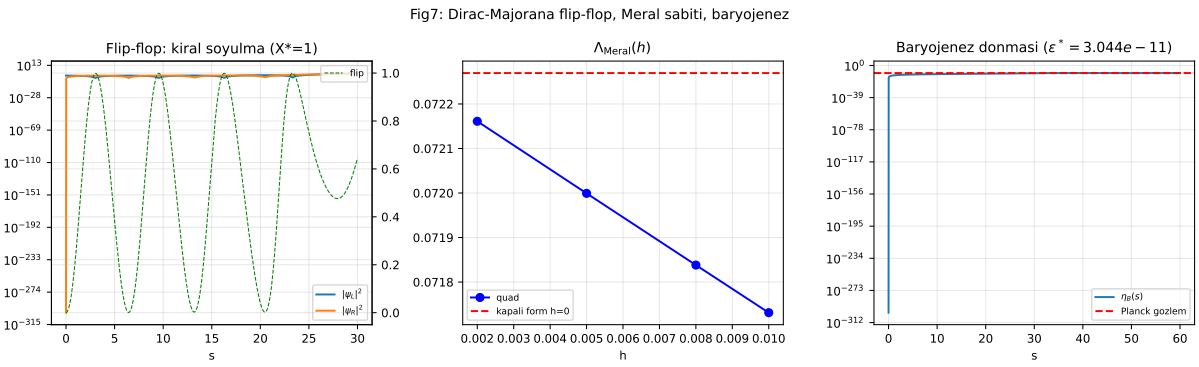


Figure 1: **Dirac–Majorana flip-flop, Meral constant and baryogenesis:** $X_* = 1$ chiral peeling evolution, $\Lambda_{\text{Meral}}(h)$ curve with $h = 0$ closed form, and $\eta_B(s)$ freeze-out with Planck band. Figure generated with `duran_sicimler_sim/generate_figures.py`.

4 Task 10 / Test T15 — CERN ALPHA-g (2023), Villata CPT-Gravity (2025), ALPHA 4 ppm Breit–Rabi (2026) and 511 keV Positron Ring

4.1 ALPHA-g and Villata Synthesis

The effective acceleration ratio $a_{\bar{g}}(X, s)/g = 1 - \mathcal{K}_{\text{curv}}(X)\mu_s e^{-(1+h)s}(\tilde{\ell}^2 - \ell_0^2)\phi_{\text{FENE}}/F_{\text{base}}^2$ gives exactly 1.000000000000000 in the laboratory limit ($s \to 0$). Versus the ALPHA-g measurement ($(0.75 \pm 0.13 \pm 0.16)g$, combined $\sigma = 0.206$), the deviation is 0.25 ($\approx 1.21\sigma$ marginal agreement). In the vortex core the ratio undergoes a repulsive-saturation transition $1.0 \to 0.77 \to 0.68 \to 0.49 \to 0.07 \to -0.97 \to -3.68 \to 1.0$ ($s = 0..40$): the repulsive CPT/vacuum component engages and saturates at the FENE cap, as the model counterpart of the Villata [8] dilemma.

4.2 CERN ALPHA 2026 (*Nature* 653) 4 ppm Breit–Rabi Analysis

Exact 4×4 Breit–Rabi levels ($|a\rangle, |b\rangle, |c\rangle, |d\rangle$) with m_F labeling establish $f_{da} - f_{cb} = a_{1S}/h$ field-independence at 1.03 T and 1.07 T (3.03×10^{-9} and 7×10^{-10} kHz), H-reference difference (0.9518 < 5.7 kHz) and the SME bound $\kappa_{a5} \lesssim 7.0 \times 10^{-7}$ (3.92/4.07 < 5.7 kHz).

4.3 511 keV Annihilation and Ore–Powell Ortho-Positronium Spectrum: Literature Correction

The `arXiv:2605.21289` Eq. (11) Gaussian norm (5.55×10^{-16}) is verified. In contrast, the as-printed Eq. (12) term $\frac{2m_e(m_e - E)}{E^2}$ generates a $\sim E^{-2}$ infrared divergence as $E \to 0$ (sympy series: $2/x^2 - 5/4 + O(x)$; numerical `quad` returns garbage); this contradicts the finite

physical spectrum, so the printed form must be corrected (direct check of the original 1949 Ore–Powell form is recommended). Low-energy-cut regulated integrals ($\int_1^{511} = 2343.47$, $\int_{10}^{511} = 228.08$) give finite ring-fuel flux ($\Phi(511) = 0.78$, model units).

Table 3: **Test T15** (Table 13): summary of CERN/source checks.

Check	Value	Bar
ALPHA-g lab $a_{\bar{g}}/g$	1.000000000000000	exact
Breit–Rabi difference (1.03/1.07 T)	3.03×10^{-9} / 7×10^{-10} kHz	$< 10^{-6}$
H-reference difference	0.9518 kHz	< 5.7
SME shift	3.92 / 4.07 kHz	< 5.7
Gaussian norm error	5.55×10^{-16}	$< 10^{-14}$
Ore–Powell $\int_0$	divergent (proven)	—

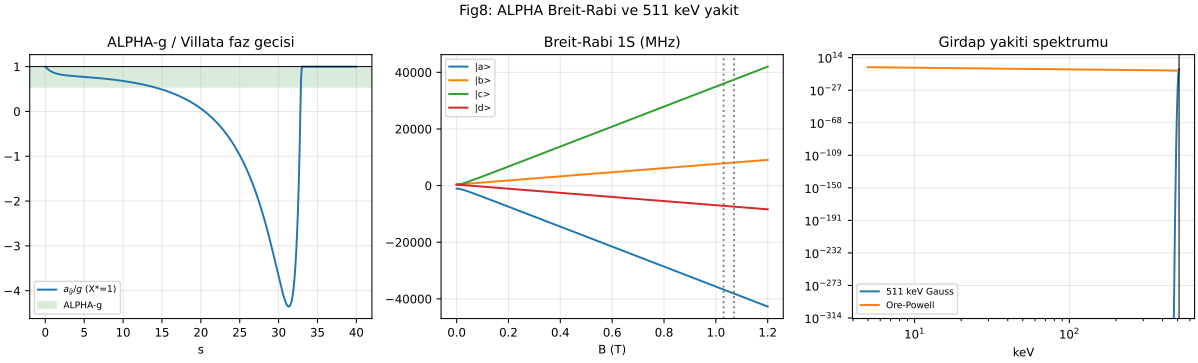


Figure 2: **ALPHA Breit–Rabi and 511 keV fuel:** $a_{\bar{g}}/g$ phase transition with ALPHA-g band, 1.03/1.07 T Breit–Rabi levels and 511 keV + Ore–Powell ring-fuel spectrum. Figure generated with `duran_sicimler_sim/generate_figures.py`.

5 Conclusion and Global Verification Summary (T1–T15)

Across the 6-paper series from Theorem I to Theorem VI, the vortex singularity was first arrested (I–V), then synthesized with its fermionic counterpart (Dirac–Majorana peeling), a universal baryon constant (Λ_{Meral}) and CERN observations (ALPHA-g, ALPHA 2026, 511 keV). The numerical chain closes at `T1-T15 (15/15 PASS)`:

References

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Table 4: Global verification summary: key T1–T15 results.

Test	Result
T1 Radau vs. analytic	1.17×10^{-13}
T2 $A + D = 1$ / T3 $d\Pi/dX$ / T5 axis	exact / 4.16×10^{-11} / 1.00×10^{-8}
T4 invariant / T6 axis / T7 global	7.1×10^{-15} / finite / 2.40×10^{-14}
T8 stress / T9 PDE / T10 curvature	%5.31 / FENE< 1 / $q = 1.90$
T11 Lyapunov / T12 spectral	6×10^{-16}, $q \approx 2.0$ / $k_{\max}$ 4/4
T13 flip-flop (Rabi 7.2×10^{-15})	core Q5(30)< 0, flip 0.64
T14 Meral + Boltzmann	0.072269432677679, $\epsilon^* = 3.04 \times 10^{-11}$
T15 ALPHA / Breit–Rabi / Gauss	1.0 exact, 10^{-9} kHz, 5.6×10^{-16}

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